

GNN (IceCube 2D) Plots (1FA006)

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The Jupyter notebook can be found in this [GitHub repository](#).

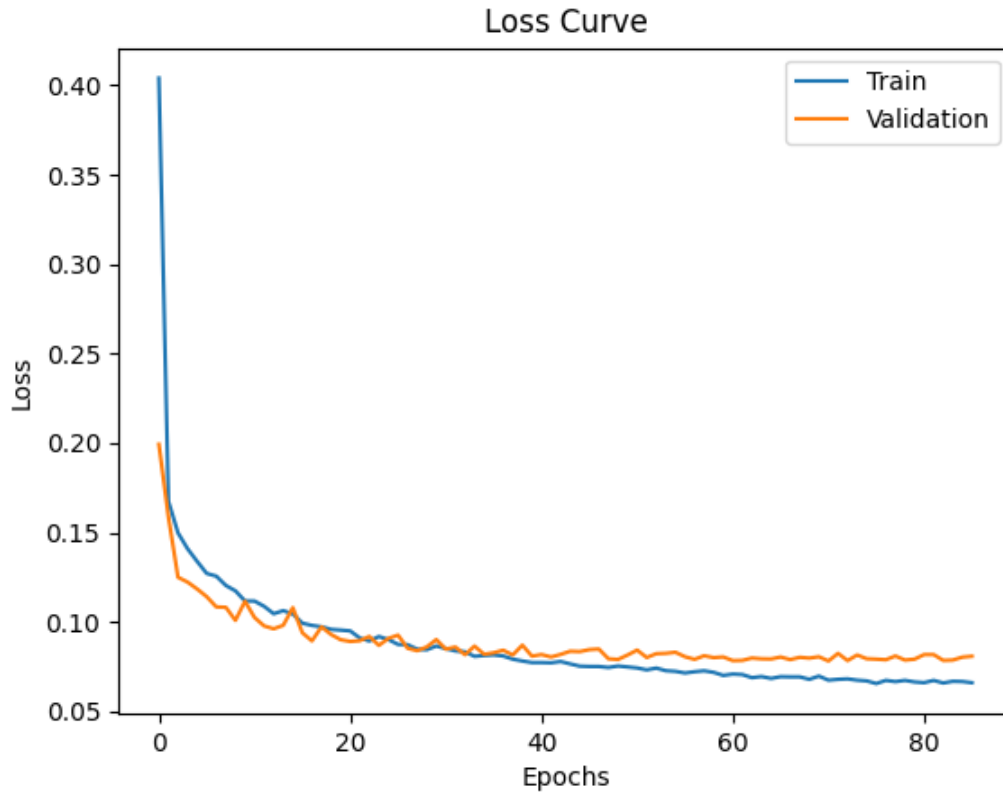


Figure 1: Loss curve of the GNN model. This shows that training is stable and saturated at around epoch 85. There is a bit of generalization error that seems to be increasing before the training early stops.

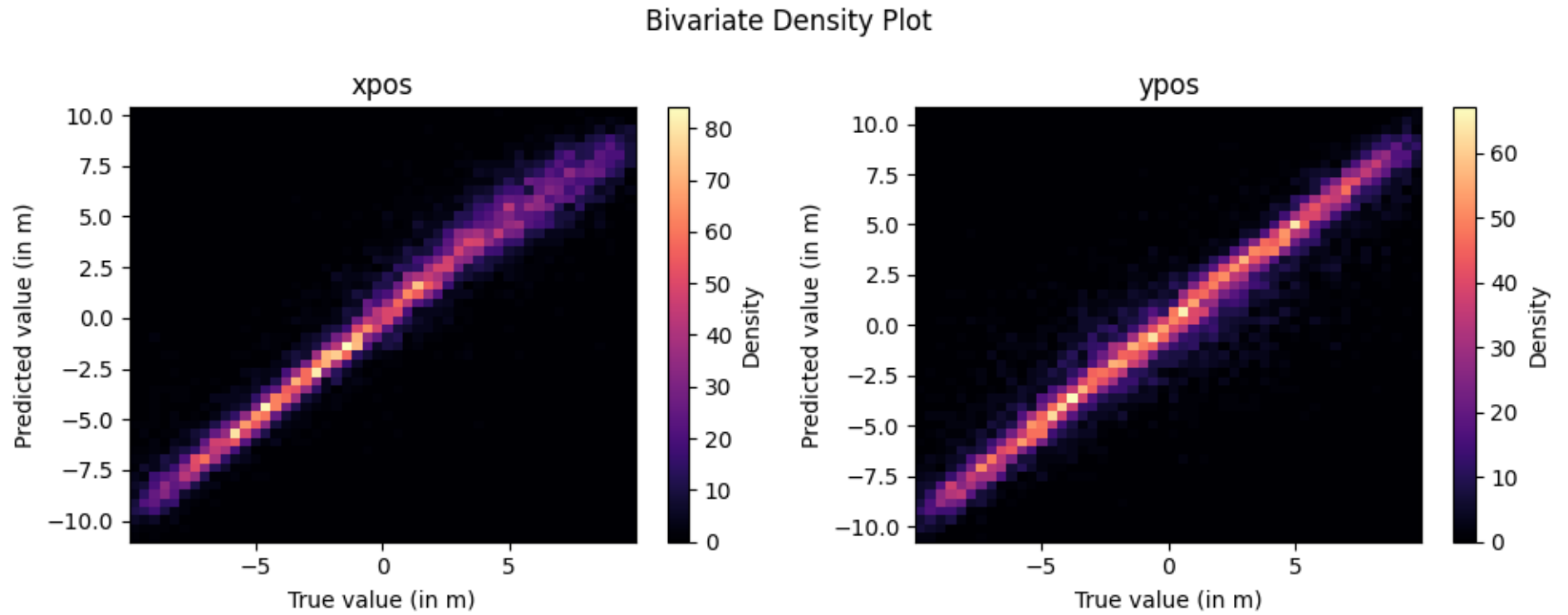


Figure 2: Predicted v/s True heatmap to show the model performance. Both the x_{pos} and y_{pos} show a clear $y = x$ trend with y_{pos} being more clearer and x_{pos} is more scattered at higher positive values.

Residual Distribution

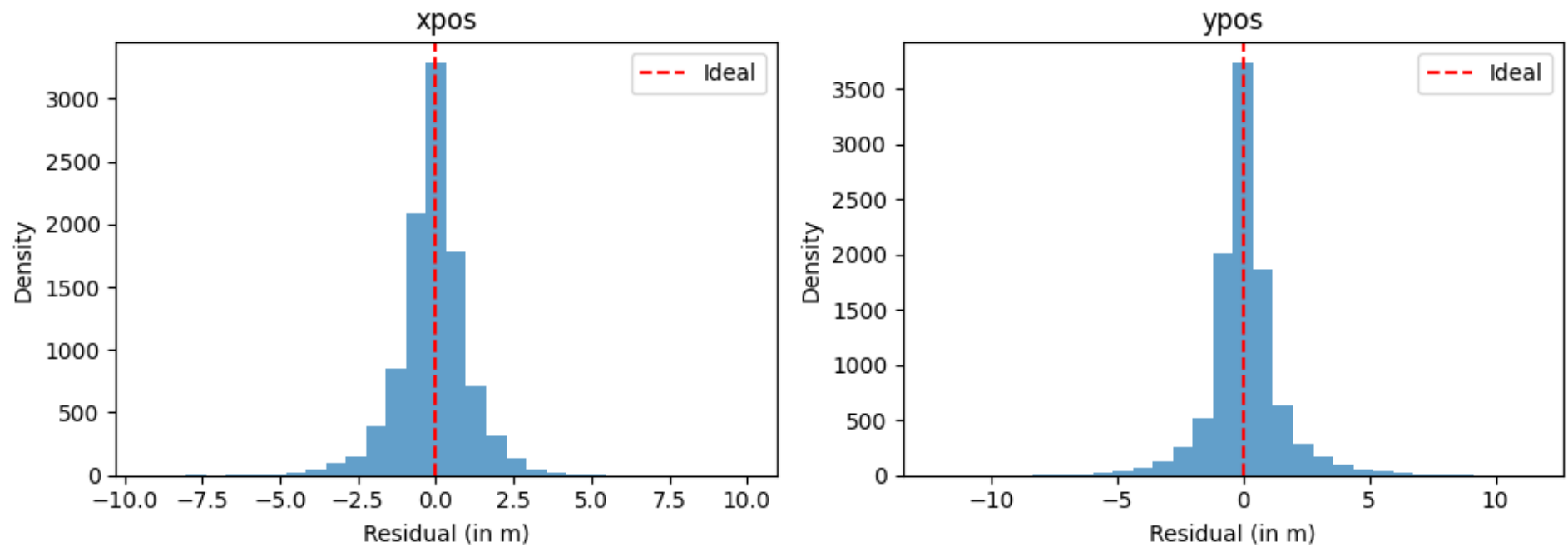


Figure 3: Residual Distributions. Both the distributions are centered around zero with a sharp peak. This means that most of the test samples have residuals (absolute value) less than 2.5m.

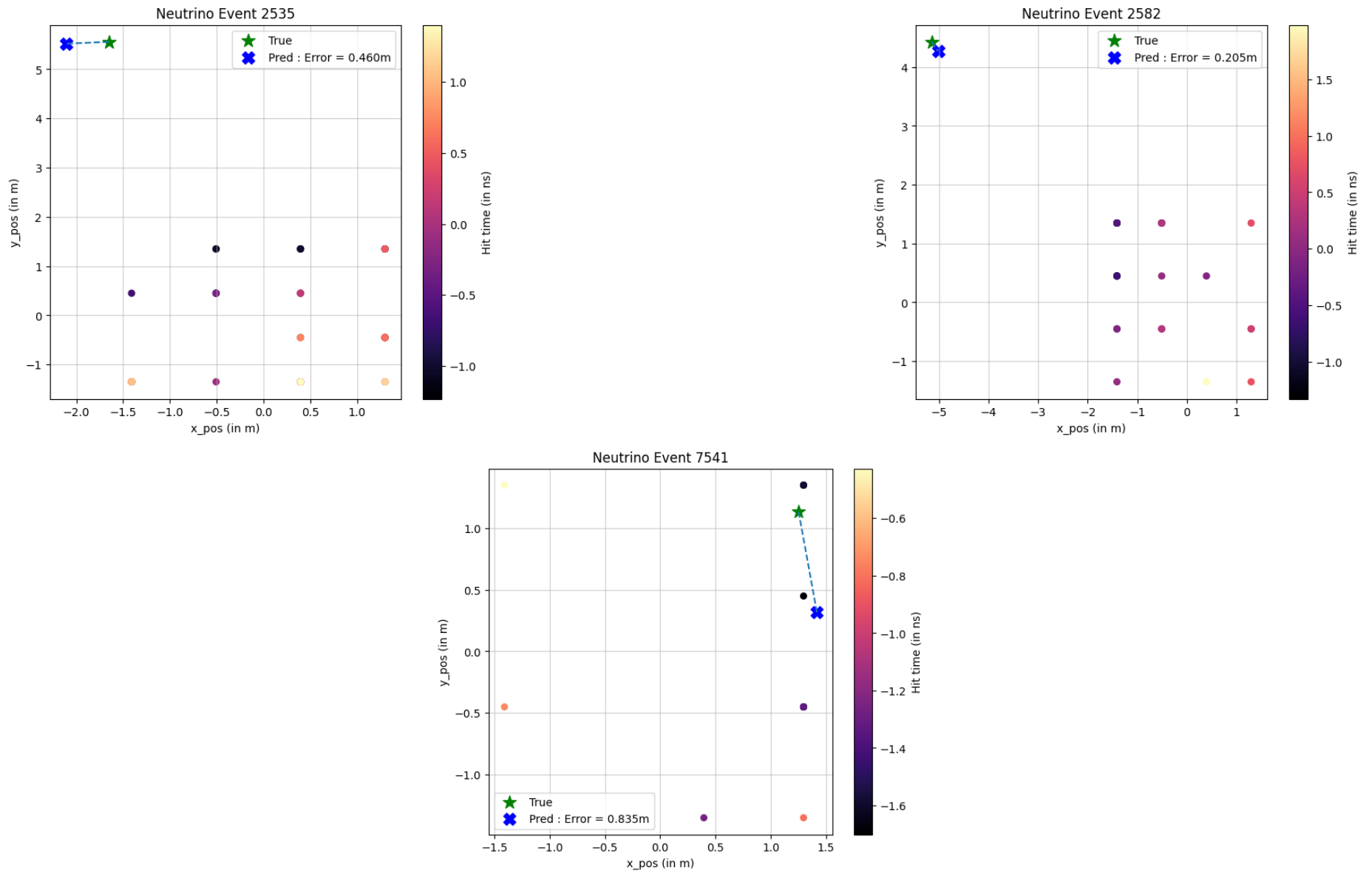


Figure 4: Examples of neutrino event positions along with the detector hit position and time, and predicted position.